

Federated Adaptive Causal Estimation with High Dimensional Covariates (FACE-HD)

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Abstract

Multi-site observational data such as electronic health records offer unprecedented opportunities for causal inference on large and heterogeneous populations, but three challenges arise in concert: privacy and regulatory constraints prohibit the sharing of patient-level data across sites, candidate confounders are high-dimensional, and covariate distributions differ markedly across sites. We propose FACE-HD, a federated adaptive causal estimation framework for high-dimensional covariates, which estimates the target-site average treatment effect (TATE) by combining a target-only doubly robust estimator with source-assisted estimators under covariate shift. In place of separate propensity score and density-ratio models at each source, we introduce an exponential-tilting calibration working model that targets the site/treatment-to-target density ratio directly through a covariate-balance moment condition; coupling this model with a generalized linear outcome regression through a two-step penalized calibrated procedure, combined with sample-splitting and two-level cross-fitting, yields pairwise estimators that are root- N consistent and asymptotically normal whenever either the outcome model or the calibration weight model is correctly specified. The pairwise estimators are aggregated with the target-only estimator through a penalized quadratic objective whose squared-gap penalty adaptively down-weights sources with detectable bias, thereby preventing negative transfer without a separate pre-screening step. A two-round federated protocol—and a pragmatic single-round variant—preserves privacy by exchanging only summary statistics across sites. Formal proofs of the asymptotic results are collected in the Supplementary Material. Simulation studies across four specification regimes illustrate the robustness and efficiency gains of FACE-HD relative to target-only, sample-size-adjusted, inverse-variance-weighted, and tilted-AIPW benchmarks.

1 Introduction

The growing availability of real-world data—electronic health records (EHRs), administrative claims, multi-center registries—has opened unprecedented possibilities for studying the comparative effectiveness of treatments in large and heterogeneous populations. Realizing that potential, however, requires navigating a constellation of methodological difficulties that rarely appear in isolation. Patient-level data typically cannot be pooled across sites because of privacy, contractual, and regulatory considerations, so any practically viable procedure must operate on summary statistics rather than on raw individual records. At the same time, modern datasets routinely harbor many potential confounders, far in excess of what classical low-dimensional causal machinery was designed to handle, and the distributions of these confounders differ across sites. Neglecting any one of federation, high-dimensional confounding, or covariate shift undermines either the feasibility or the validity of the resulting inference [Tong et al., 2025].

A line of recent work has begun to address subsets of these issues. Tong et al. [2025] developed DisC2o-HD, a distributed procedure that calibrates penalized propensity score and outcome models to attain covariate balance and doubly robust inference under covariate shift, although the estimand it targets is the site-specific treatment effect rather than its transport to a designated target population. Han et al. [2023] proposed a multiply robust federated estimator that draws on transfer-learning devices to combine information across sites while preserving privacy, but the framework is fundamentally low-dimensional and requires separate modeling of the propensity score and the density ratio, which doubles the nuisance dimension in the high-dimensional regime. The Federated Adaptive Causal Estimation (FACE) framework of Han et al. [2025] closes the transport gap by combining target-site and source-site doubly robust estimators through data-adaptive aggregation weights that perform source selection; its validity, however, hinges on pre-specification of the confounder set. In EHR-scale applications that

presupposition is difficult to sustain: the relevant confounders must themselves be selected from a large candidate pool, and standard variable-selection devices introduce a bias that destroys the asymptotic linearity of the source-site augmentation terms and, with them, the aggregation step at the heart of FACE. In a parallel strand of the literature, the calibrated ℓ_1 -penalized estimators of Smucler et al. [2019] and Tan [2020] deliver doubly robust inference for average treatment effects in high dimensions, but only in a single-site setting. A unified framework that simultaneously accommodates federation, high-dimensional confounding, and covariate shift has so far been missing.

This paper develops such a framework. The proposed FACE-HD procedure extends FACE to the high-dimensional regime along three coordinated lines of design. The first is the replacement of separate propensity score and density-ratio models by a single exponential-tilting calibration working model whose regularized convex loss enforces a covariate-balance moment condition at the target population. The calibration parameters play the role of a combined propensity–density-ratio nuisance, and the induced weight function admits a stable, strictly positive form that is convenient in high dimensions. The second is the construction of cross-calibrated penalized losses for the outcome model and the calibration weight model in the spirit of Smucler et al. [2019] and Tan [2020], embedded in a two-level cross-fitting scheme adapted from Hou et al. [2025] and Chernozhukov et al. [2018] to enforce Neyman orthogonality between the two nuisances. The resulting pairwise estimators are $\sqrt{N_{\text{all}}}$ -consistent and asymptotically normal whenever either the outcome model or the calibration weight model is correctly specified, and are computed in a federated manner that exposes only summary statistics. The third is the aggregation of the pairwise estimators with the target-only estimator through a penalized quadratic objective, whose penalty is scaled by the squared gap between each pairwise estimator and the target-only estimator. Under a separation condition on biased sources, this penalty drives the weights of substantially biased sources to zero while retaining the informative sources at their variance-optimal weights. Selection consistency of the weights and oracle efficiency of the aggregated estimator are formally established in the Supplementary Material. The procedure is naturally compatible with federation: we describe a two-round protocol that transmits only gradient-type summary statistics between the target and the source sites, together with a single-round pragmatic variant that further reduces communication cost.

The remainder of the paper is organized as follows. Section 2 develops the problem setting, the exponential-tilting calibration weights, the GLM outcome models, and the resulting pairwise doubly robust estimators. Section 3 derives the cross-calibrated losses and states the two-step estimation procedure. Section 4 presents the influence-function representation, the variance and covariance components, the penalized aggregation objective, a dedicated treatment of negative transfer, and a summary of the regularity conditions required for the asymptotic results. Section 5.1 describes the two-round federated algorithm, while Section 6 discusses cross-validation for the nuisance models together with a data-adaptive grid for the aggregation penalty. Section 7 reports the simulation study, and Section 8 concludes. One-round variants, two-level cross-fitting details, and full proofs are deferred to the appendices and the Supplementary Material.

2 Framework for Multi-Site Causal Inference

2.1 Problem Setting and Notation

We address a causal inference problem involving a single target site (t) and multiple source sites ($\mathcal{S} = \{s_1, s_2, \dots, s_K\}$). Our primary objective is to estimate the average treatment effect (ATE) at the target site by leveraging data from both the target site and auxiliary information from the source sites. We define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the ATE

$$\tau := \mu^1 - \mu^0.$$

Throughout, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. For exposition, we derive the nuisance losses, algorithms, and asymptotic results for μ^1 ; the corresponding statements for μ^0 follow by symmetry (replace treatment level 1 by 0 and use the corresponding nuisance parameters), and the ATE estimator is obtained by differencing:

$$\hat{\tau} := \hat{\mu}^1 - \hat{\mu}^0.$$

Basic Notation:

- $R \in \{t, s_1, \dots, s_K\}$: A site membership indicator.

- $\mathbf{X}$: A vector of raw covariates. We define $\phi(\mathbf{X})$ as a basis expansion (e.g., polynomials, splines) of the covariates to allow for flexible modeling.
- $A \in \{0, 1\}$: A binary treatment indicator.
- Y : The outcome of interest. $Y(1)$ and $Y(0)$ are the potential outcomes.
- $\mathcal{D}_t, \mathcal{D}_{s_j}$: Data from the target site t and source site s_j .
- N_t, N_{s_j} : Sample sizes at the target and source sites.
- $N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j}$: Total sample size across all sites.
- $\tilde{E}_r(\cdot)$: Denotes the empirical average over data from site r (e.g., $r = t$ or $r = s_j$).

2.2 Generalized Linear Model Framework

2.2.1 Exponential-Tilting Calibration Weights for Site/Treatment Reweighting

For each source site $s_j \in \mathcal{S}$ and treatment level $a \in \{0, 1\}$, we construct *calibration weights* that reweight the $(R = s_j, A = a)$ group to match the target covariate distribution. We use an *exponential tilting* (log-linear) *working model* for the tilting function

$$r_{s_j,a}(\mathbf{X}; \boldsymbol{\gamma}_{s_j,a}) := \exp\{\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,a}\}. \quad (1)$$

This form is convenient because it yields convex calibration losses and strictly positive weights. We do *not* interpret (1) as a literal probabilistic model for $P(R, A \mid \mathbf{X})$; instead, the key requirement for our theory is that the implied weights satisfy covariate balance (Equation (3)) at the population target. Crucially, for the pairwise DR estimator $\hat{\mu}_{t,s_j}^1$, only the single parameter vector $\boldsymbol{\gamma}_{s_j,1}$ (for the treated group at source s_j) is required.

Given (1), we define the (unnormalized) calibration weight for source-site treated observations as

$$w(\mathbf{X}; \boldsymbol{\gamma}_{s_j,1}) := I(A = 1) \exp\{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,1}\} = \frac{I(A = 1)}{e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j,1}}}. \quad (2)$$

Because $\phi(\mathbf{X})$ includes an intercept, any multiplicative normalizing constant (equivalently, an additive constant in the log-tilt) is absorbed into the intercept component of $\boldsymbol{\gamma}_{s_j,1}$. Consequently, when $\boldsymbol{\gamma}_{s_j,1}^*$ is the population solution of the calibration/balancing equation (the initialization loss FOC in (12) and the calibrated loss FOC in (50)), the operational weight satisfies the *balancing property*: for every integrable function h ,

$$\mathbb{E}_{s_j}[w(\mathbf{X}; \boldsymbol{\gamma}_{s_j,1}^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})], \quad (3)$$

In general, the identity (3) for *all* integrable h corresponds to the ideal (population) density-ratio weighting relationship. In our implementation and high-dimensional analysis, we enforce balance over a rich but finite-dimensional function class, typically the linear span of the chosen features $h(\mathbf{X}) \in \text{span}\{\phi_1(\mathbf{X}), \dots, \phi_p(\mathbf{X})\}$ (including an intercept). The corresponding “weight-model correctness” condition in the appendix should be interpreted relative to this function class, which is the key identity underlying the doubly robust estimator in Section 2.3.

Remark 1 (Calibration weights vs. full probabilistic model). The exponential-tilting form (1) can be motivated by a multinomial logistic model for $(R, A \mid \mathbf{X})$ (with $R = t$ as a reference category), in which case the implied weights approximate inverse odds of site/treatment membership. In this paper, however, we treat (1) as a *calibration working model*: $\boldsymbol{\gamma}_{s_j,a}$ is defined by the (regularized) calibration loss, and the substantive requirement is the resulting covariate-balance moment condition (3).

2.2.2 GLM-Based Outcome Models

We use generalized linear models for the potential outcomes. Let $\psi(\cdot; \cdot)$ be a GLM response function with a canonical link, mapping features to the conditional mean of the outcome.

$$m^a(\mathbf{X}; \boldsymbol{\alpha}^a) = \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}^a), \quad a \in \{0, 1\} \quad (4)$$

2.3 Doubly Robust Estimators

The doubly robust estimators combine outcome modeling and inverse probability weighting. The estimator for μ^1 assisted by source site s_j is:

$$\hat{\mu}_{t,s_j}^1 = \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) + \tilde{E}_{s_j} \left(\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \tilde{\gamma}_{s_j,1}}} (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \right) \quad (5)$$

The estimator for the control group, $\hat{\mu}_{t,s_j}^0$, is defined symmetrically by using the nuisance parameters $(\alpha^0, \gamma_{s_j,0})$ and replacing $I(A=1)$ by $I(A=0)$ in the weighted residual term:

$$\hat{\mu}_{t,s_j}^0 = \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^0)) + \tilde{E}_{s_j} \left(\frac{I(A=0)}{e^{\phi(\mathbf{X})^T \tilde{\gamma}_{s_j,0}}} (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^0)) \right). \quad (6)$$

Finally, the (pairwise) ATE estimator is $\hat{\tau}_{t,s_j} := \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$, and the target-only ATE estimator is $\hat{\tau}_{ot} := \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$.

Target-site propensity score. The target-only estimator $\hat{\mu}_{ot}^1$ is a standard AIPW estimator on the target data alone and requires the *target-site propensity score*,

$$\pi_{ot}(\mathbf{X}) := P(A=1 \mid R=t, \mathbf{X}), \quad (7)$$

which models treatment assignment *within* the target population. This is conceptually and operationally distinct from the cross-site calibration/tilting parameters $\gamma_{s_j,a}$ defined in Section 2.2.1: π_{ot} is estimated locally at the target site (e.g., via ℓ_1 -penalized logistic regression on $(A, \mathbf{X})$ pairs from $\mathcal{D}_t$), and does not enter the source-assisted estimator $\hat{\mu}_{t,s_j}^1$ or its loss functions.

3 Parameter Estimation and Loss Functions

3.1 The Coupling Challenge in Federated Settings

The outcome model parameters α_{t,s_j}^a and the site/treatment calibration parameters $\gamma_{s_j,a}$ enter the doubly robust estimator (5) in a coupled manner: the optimal target of α depends on the calibration weights induced by γ , while the optimal target of γ depends on derivatives of the outcome mean at α . A joint solution of the two nuisance problems is, in principle, available at a single site, but in a federated setting—where neither party can examine the other’s raw data—such joint optimization would require an excessive number of communication rounds and would entangle the convergence of the two components. We therefore pursue a decoupled, alternating estimation strategy that respects the communication constraints and nonetheless preserves the Neyman orthogonality property required for root- N inference.

3.2 Derivation of Loss Functions

We derive the cross-calibrated loss functions by requiring that the estimating equations for the nuisances coincide with the efficient score equations implied by the influence-function expansion of the pairwise estimator (5). This construction guarantees that the leading-order contribution of the nuisance estimation error to $\hat{\mu}_{t,s_j}^1 - \mu^1$ vanishes at the population level, leaving a remainder that is second-order in the nuisance errors and therefore asymptotically negligible under standard high-dimensional rates.

Step 1: Influence Function Decomposition. The asymptotic error of $\hat{\mu}_{t,s_j}^1$ can be decomposed into terms related to the estimation errors of the parameters α and γ :

$$\hat{\mu}_{t,s_j}^1 - \mu_{t,s_j}^1 = (\hat{\alpha}_{t,s_j}^1 - \bar{\alpha}_{t,s_j}^1)^T \mathbf{U}_\alpha + (\hat{\gamma}_{s_j,1} - \bar{\gamma}_{s_j,1})^T \mathbf{U}_\gamma + \text{h.o.t.} \quad (8)$$

where $\bar{\alpha}_{t,s_j}^1$ and $\bar{\gamma}_{s_j,1}$ denote the *population targets* (probability limits / pseudo-true parameters) associated with the nuisance estimation step. Under correct specification of the corresponding nuisance model, these coincide with the true parameters. The vectors $\mathbf{U}_\alpha$ and $\mathbf{U}_\gamma$ are the score vectors:

$$\mathbf{U}_\alpha = \tilde{E}_t \left[\nabla_\alpha \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1) \right] - \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \tilde{\gamma}_{s_j,1}}} \nabla_\alpha \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1) \right] \quad (9)$$

$$\mathbf{U}_\gamma = \tilde{E}_{s_j} \left[-\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \tilde{\gamma}_{s_j,1}}} \phi(\mathbf{X}) (Y - \psi(\phi(\mathbf{X}); \bar{\alpha}_{t,s_j}^1)) \right] \quad (10)$$

Step 2: Cross-Calibration Principle. For the remainder in (8) to be second-order in the nuisance errors (Neyman orthogonality / debiasing), the estimators must be constructed so that $\mathbf{U}_\alpha = \mathbf{0}$ at $\gamma = \hat{\gamma}$ and $\mathbf{U}_\gamma = \mathbf{0}$ at $\alpha = \hat{\alpha}$. The cross-calibration strategy achieves this by constructing the loss for γ so that its first-order condition is the score equation $\mathbf{U}_\alpha = \mathbf{0}$, and symmetrically for α .

Step 3: Calibrated Loss for γ and Verification. We use the calibrated exponential-tilting loss in Eq. (50). To verify the cross-calibration, take the gradient with respect to γ (setting aside the ℓ_1 subdifferential):

$$\nabla_\gamma \ell = \tilde{E}_t[\nabla_\alpha \psi(\phi(\mathbf{X}); \bar{\alpha})] - \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \gamma}} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \bar{\alpha})) \phi(\mathbf{X}) \right].$$

For a canonical-link GLM, $\nabla_\alpha \psi(\phi(\mathbf{X}); \alpha) = \psi'(\phi(\mathbf{X})^T \alpha) \phi(\mathbf{X})$, so under asymptotically inactive truncation the two terms become

$$\nabla_\gamma \ell = \tilde{E}_t[\psi'(\phi^T \bar{\alpha}) \phi(\mathbf{X})] - \tilde{E}_{s_j}[w(\mathbf{X}; \gamma) \psi'(\phi^T \bar{\alpha}) \phi(\mathbf{X})],$$

which matches the score vector $\mathbf{U}_\alpha$ in (9) up to an asymptotically negligible truncation term. *Hence the population first-order condition of the calibrated γ -loss enforces the same first-order orthogonality equation $\mathbf{U}_\alpha = \mathbf{0}$.* Moreover, the calibrated loss (50) is strictly convex in γ (because $t \mapsto e^{-t}$ is convex), guaranteeing a unique minimizer under standard regularity.

Step 4: Calibrated Loss for α and Verification. The outcome model is fitted by minimizing the calibrated weighted GLM negative log-likelihood in Eq. (51), given a plug-in site/treatment model $\hat{\gamma}$. where $w(\mathbf{X}; \hat{\gamma}) = I(A=1)/e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j,1}}$ is the exponential-tilting calibration weight from (2), and $\Psi(\cdot)$ is the log-partition (cumulant) function of the GLM. The gradient is

$$\nabla_\alpha \ell = \tilde{E}_{s_j}[w_\tau(\mathbf{X}; \hat{\gamma}) (-Y + \psi(\phi(\mathbf{X})^T \alpha)) \phi(\mathbf{X})],$$

and under asymptotically inactive truncation this matches $-\mathbf{U}_\gamma(\alpha, \hat{\gamma})$ at first order, so the first-order condition $\nabla_\alpha \ell = \mathbf{0}$ enforces $\mathbf{U}_\gamma = \mathbf{0}$.

Cross-calibration summary. Steps 3–4 establish the cross-calibration: minimizing the γ -loss enforces $\mathbf{U}_\alpha = \mathbf{0}$, and minimizing the α -loss enforces $\mathbf{U}_\gamma = \mathbf{0}$. Together, the first-order terms in the remainder (8) vanish at the population level, leaving a second-order error $O_p(\|\hat{\alpha} - \alpha^*\| \cdot \|\hat{\gamma} - \gamma^*\|)$ that is negligible under standard high-dimensional rates; see the appendix proofs for details.

3.3 Two-Step Estimation Procedure

To resolve the parameter coupling described in Section 3 under the communication constraints of a federated setting, we proceed by an alternating two-step estimation strategy in which each of the two nuisance models is estimated once using an initial, non-calibrated plug-in for the other, and once using the resulting cross-calibrated loss.

Step 1: Initial Parameter Estimation (Iteration 0). In the first step we obtain initial estimates that are consistent under the corresponding model but do not yet satisfy the cross-calibration moment conditions; these serve as plug-ins in the subsequent calibration step.

Initial Outcome Model: A simple regularized regression on source site data provides a starting point $\hat{\alpha}_{t,s_j}^{1,(0)}$. Let ℓ_{GLM} be the standard loss for the chosen GLM (e.g., squared error for linear, log-loss for logistic).

$$\ell(\alpha_{t,s_j}^1) = \tilde{E}_{s_j} \left[I(A=1) \ell_{GLM}(Y, \psi(\phi(\mathbf{X}); \alpha_{t,s_j}^1)) \right] + \lambda_\alpha \|\alpha_{t,s_j}^1\|_1 \quad (11)$$

Initial Site/Treatment Model: We initialize $\hat{\gamma}_{s_j,a}^{(0)}$ by minimizing the following regularized exponential tilting loss, which targets the density ratio weights without depending on the outcome model, providing a robust starting point.

$$\ell(\gamma_{s_j,1}) = \left(\tilde{E}_t[\phi(\mathbf{X})] \right)^T \gamma_{s_j,1} + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma_{s_j,1}} \right] + \lambda_\gamma \|\gamma_{s_j,1}\|_1 \quad (12)$$

Step 2: Calibrated Parameter Estimation (Iteration 1). Using the initial estimates, we perform one round of alternating minimization.

1. **Calibrated Site/Treatment Model:** Obtain $\hat{\gamma}_{s_j,1}^{(1)}$ by minimizing the calibrated loss function (50) with the initial $\hat{\alpha}_{t,s_j}^{1,(0)}$ plugged in.
2. **Calibrated Outcome Model:** Obtain $\hat{\alpha}_{t,s_j}^{1,(1)}$ by minimizing the calibrated loss function (51) using the initial site/treatment model $\hat{\gamma}_{s_j,1}^{(0)}$ as the plug-in weight model (this plug-in order is intentional in the two-level cross-fitting construction and does not affect first-order asymptotics under the alignment condition stated in the appendix).

The final parameters used are $\hat{\alpha}_{t,s_j}^1 := \hat{\alpha}_{t,s_j}^{1,(1)}$ and $\hat{\gamma}_{s_j,1} := \hat{\gamma}_{s_j,1}^{(1)}$.

4 Asymptotic Properties, Aggregation, and Inference

4.1 Influence Functions and Variance Components

Confidence intervals and optimal aggregation weights both rely on the influence-function representation of the pairwise estimators introduced in Section 2.3. The asymptotic variance of the aggregated estimator decomposes into variance and covariance components of these influence functions, and consistent plug-in estimators of these components are what ultimately drive the selection behavior of the aggregation step.

Let $\hat{\mu}_{ot}^1$ be the target-only estimator and $\hat{\mu}_{t,s_j}^1$ be the source-assisted estimator from site s_j .

4.1.1 Specific Influence Functions

The empirical influence functions are estimated as follows:

Target-Only Estimator IF ($\hat{\varphi}_{ot,i}$): Using the target-site propensity score $\hat{\pi}_{ot}(\mathbf{X})$ defined in (7) and a target-only outcome model $\hat{\alpha}_{ot}^1$:

$$\hat{\varphi}_{ot,i} = \frac{A_i Y_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{ot}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{ot}^1) - \hat{\mu}_{ot}^1 \quad (13)$$

Target-Site Component IF ($\hat{\zeta}_{t,s_j,i}$):

$$\hat{\zeta}_{t,s_j,i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1) - \tilde{E}_t(\psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \quad (14)$$

Source-Site Component IF ($\hat{\xi}_{t,s_j,i}$): Using the calibration weight $w(\mathbf{X}; \gamma)$ from (2):

$$\hat{\xi}_{t,s_j,i} = \frac{A_i}{e^{\phi(\mathbf{X}_i)^T \hat{\gamma}_{s_j,1}}} (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1)) - \hat{\delta}_{t,s_j}^1 \quad (15)$$

where $\hat{\delta}_{t,s_j}^1 = \tilde{E}_{s_j} \left(\frac{A}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j,1}}} (Y - \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \right)$.

4.1.2 Variance Components

From these influence functions, we can estimate the necessary variance and covariance components.

$$\hat{V}_{ot} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot})^2 \quad (16)$$

$$\hat{V}_{t,s_j} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i} - \bar{\zeta}_{t,s_j})^2 \quad (17)$$

$$\hat{V}_{s_j} = \frac{1}{N_{s_j}} \sum_{i \in \mathcal{D}_{s_j}} (\hat{\xi}_{t,s_j,i} - \bar{\xi}_{t,s_j})^2 \quad (18)$$

$$\hat{C}_{ot,s_j} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot})(\hat{\zeta}_{t,s_j,i} - \bar{\zeta}_{t,s_j}) \quad (19)$$

Additionally, we must compute cross-site covariance terms to account for the fact that all source-assisted estimators use the same target site data:

$$\hat{C}_{t,s_j,s_k} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i} - \bar{\hat{\zeta}}_{t,s_j})(\hat{\zeta}_{t,s_k,i} - \bar{\hat{\zeta}}_{t,s_k}) \quad \text{for } j \neq k \quad (20)$$

Ignoring these cross-site correlations would lead to a systematic underestimation of the aggregated estimator's variance and incorrect confidence intervals; see also the analogous shared-target covariance correction in FACE [Han et al., 2025].

Finite-sample numerical note. In finite samples, the plug-in variance/covariance components (especially the shared-target covariance terms) can yield a quadratic objective in (22) that is only approximately convex due to estimation noise. Our implementation therefore symmetrizes the estimated covariance inputs and applies a minimal ridge regularization when needed to keep the weight optimization numerically well-posed; this ridge vanishes asymptotically under the regularity conditions in the appendix.

4.2 Optimal Aggregation of Estimators

The target site combines its own estimator, $\hat{\mu}_{ot}^1$, with the estimators received from all source sites, $\{\hat{\mu}_{t,s_j}^1\}_{j=1}^K$. The final aggregated estimator is a weighted combination:

$$\hat{\mu}_{agg}^1 = (1 - \sum_{j=1}^K \eta_j) \hat{\mu}_{ot}^1 + \sum_{j=1}^K \eta_j \hat{\mu}_{t,s_j}^1 \quad (21)$$

The optimal weights $\boldsymbol{\eta}^* = (\eta_1^*, \dots, \eta_K^*)$ are found by minimizing the asymptotic variance of $\hat{\mu}_{agg}^1$, with an added penalty term to regularize the weights and perform source selection. This leads to the following quadratic optimization problem:

$$\begin{aligned} \min_{\boldsymbol{\eta}} \quad & \frac{1}{N_t} \left(1 - \sum_{j=1}^K \eta_j\right)^2 \hat{V}_{ot} + \sum_{j=1}^K \frac{\eta_j^2}{N_t} \hat{V}_{t,s_j} + \sum_{j=1}^K \frac{\eta_j^2}{N_{s_j}} \hat{V}_{s_j} \\ & + \frac{2}{N_t} \sum_{j=1}^K \eta_j \left(1 - \sum_{k=1}^K \eta_k\right) \hat{C}_{ot,s_j} + \frac{2}{N_t} \sum_{j < k} \eta_j \eta_k \hat{C}_{t,s_j,s_k} \\ & + \lambda \sum_{j=1}^K |\eta_j| \left(\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1\right)^2 \end{aligned} \quad (22)$$

Remark on Scaling. Equation (22) is written in terms of an estimated variance $\widehat{\text{Var}}(\hat{\mu}_{agg}^1)$. For asymptotic analysis of $\sqrt{N_{\text{all}}}(\hat{\mu}_{agg}^1 - \mu^1)$, one can equivalently work with the estimated asymptotic variance $N_{\text{all}} \cdot \widehat{\text{Var}}(\hat{\mu}_{agg}^1)$, which multiplies the variance and covariance terms in (22) by N_{all} . This scaling can be absorbed into the tuning of λ and is used in the appendix proofs.

ATE aggregation. Define $\hat{\mu}_{agg}^0$ by applying the same aggregation construction to $\{\hat{\mu}_{t,s_j}^0\}_{j=1}^K$ (with the corresponding variance/covariance components for the control arm), and set $\hat{\tau}_{agg} := \hat{\mu}_{agg}^1 - \hat{\mu}_{agg}^0$. The appendix gives the asymptotic linearity/normality argument; inference for τ follows by standard linear combination (difference) of the two arm-specific influence function representations.

4.3 Preventing Negative Transfer via Penalized Aggregation

A federated estimator that pools information across heterogeneous sources is always exposed to the possibility of *negative transfer*: when the conditional outcome distribution or the covariate-balance relation at a source site deviates substantially from its target-site counterpart, the corresponding pairwise estimator $\hat{\mu}_{t,s_j}^1$ converges to a limit different from μ^1 , and including it in the aggregation injects asymptotic bias rather than efficiency. A procedure that behaves well across a range of data-generating mechanisms must therefore be able to recognize such sources and to down-weight them automatically.

The penalty term $\lambda \sum_{j=1}^K |\eta_j| (\hat{\mu}_{ot}^1 - \hat{\mu}_{t,s_j}^1)^2$ in the aggregation objective (22) is engineered for precisely this purpose. Sources whose pairwise estimators remain close to the target-only estimator $\hat{\mu}_{ot}^1$ incur only negligible penalty and are retained at weights that minimize the asymptotic variance of the aggregated estimator; sources whose pairwise estimators are detectably biased pay a penalty that grows with the squared gap and are driven to weight zero. In this sense the mechanism can be viewed as a data-adaptive analog of the site screening performed heuristically in applied multi-site studies, but embedded directly into the estimating equation so that the bias and variance trade-offs are coherently resolved.

To formalize the behavior, let $\mathcal{S}^* := \{j : \lim_N \hat{\mu}_{t,s_j}^1 = \mu^1\}$ denote the set of *informative* source sites, that is, those whose pairwise estimators are asymptotically unbiased for the target parameter. Under a separation (β -min) condition requiring the bias of any non-informative source to be detectable at the penalty scale, together with the regularity conditions summarized below, the penalized weights $\hat{\eta}$ satisfy the selection property $\hat{\eta}_j = 0$ with probability approaching one for every $j \notin \mathcal{S}^*$; correspondingly, the aggregated estimator attains the oracle asymptotic variance restricted to $\mathcal{S}^*$ and is, in particular, asymptotically at least as efficient as the target-only estimator. Formal statements and proofs of these selection and efficiency results are given in Lemma B.5 and Theorem B.7 of the Supplementary Material. When biased sources are only weakly biased—that is, when the separation scale shrinks with N —exact selection may fail, but the aggregated estimator remains asymptotically linear for any deterministic weight configuration and the incurred bias can be controlled at rate λ/N_{all} .

Operationally, the penalty λ is tuned on a data-adaptive logarithmic grid whose center λ_{nat} is set to the median of the source-specific transitions $\lambda_j = \sigma_j^2 / \max(d_j^2, \sigma_j^2)$, where σ_j^2 and d_j^2 denote the estimated pairwise variance and the squared gap to the target-only estimator (Section 6.2); this calibration ensures that λ scales correctly relative to the estimated bias and variance magnitudes, so that well-aligned sources are retained and substantially biased sources are discarded without manual pre-screening.

4.4 Theoretical Regularity Conditions (Summary)

The detailed proofs are relegated to the Supplementary Material. The pairwise asymptotic normality of the cross-fitted doubly robust estimator $\hat{\mu}_{t,s}^{1,\text{CF}}$ is established in Theorem A.3, and the asymptotic normality and oracle efficiency of the aggregated FACE-HD estimator are established in Theorem B.7; identification and double robustness are established in Lemma A.1, and the source-selection consistency underlying the discussion of negative transfer in Section 4.3 is established in Lemma B.5. In the present section we summarize the main technical conditions that these results rely on and clarify a few points that are likely to be questioned by readers.

- **High-dimensional nuisance rates under calibrated losses.** The calibrated nuisance estimators $\hat{\alpha}$ and $\hat{\gamma}$ are ℓ_1 -penalized M-estimators. Under standard sub-Gaussian design/tail conditions and Restricted Strong Convexity (RSC), they satisfy $\|\hat{\beta} - \beta^*\|_2 = O_p(\sqrt{s \log(p)/N})$ and $\|\hat{\beta} - \beta^*\|_1 = O_p(s \sqrt{\log(p)/N})$ for $\beta \in \{\alpha, \gamma\}$; see Tan [2020], Negahban et al. [2012], Bühlmann and Van De Geer [2011].
- **Remainder control via calibration + cross-fitting.** The influence-function expansion has a remainder that is second-order in the nuisance errors because the calibrated losses enforce population score/orthogonality conditions. Cross-fitting ensures the evaluation fold is independent of the nuisance training sample, simplifying the empirical-process bounds (see also the general cross-fitting/orthogonality perspective in Chernozhukov et al. [2018]).
- **Variance/covariance estimation with shared target data.** Because all source-assisted estimators use the same target sample, the cross-site covariance terms $\hat{C}_{t,s_j,s_k}$ in (22) are required for a consistent variance estimate. These are still consistent plug-in estimators because they are ordinary sample averages over the i.i.d. target observations (with cross-fitted nuisances).
- **Source selection and the “separation” assumption.** The support recovery guarantee for $\hat{\eta}$ uses a β -min type condition: biased sites must differ from the target estimand by at least a detectable amount so that the penalty dominates and drives $\hat{\eta}_j \rightarrow 0$. More plainly: consistency of the aggregated estimator relies on the aggregation putting (asymptotically) negligible weight on non-transportable sources. If all sources have non-vanishing systematic bias for the target estimand and yet receive non-negligible weights, then $\hat{\mu}_{\text{agg}}^1$ will inherit bias (it targets a biased weighted mixture). Under weak bias/local alternatives (bias shrinking with N), exact selection consistency may fail; inference remains valid as long as any included weakly biased sources contribute $o_p(N^{-1/2})$ bias at the target scale.

- **Bounded weights and clipping.** In practice one may clip extreme linear predictors during nuisance *optimization* for numerical stability; this corresponds to using the truncation map $\mathcal{T}(\cdot)$ inside the calibrated nuisance losses (controlled by M_τ). Our default inference formulas use the *untruncated* DR estimating equation (equivalently, $M_{\tau, \text{inference}} = \infty$ in the software). If one also applies truncation/clipping inside the final DR estimating equation, then the influence function changes in finite samples; the first-order limit theory is unchanged when truncation is asymptotically inactive (e.g., clipping probability $o(N^{-1})$ at the population targets), otherwise a clipping-induced bias term must be controlled (see Tan [2020] for related discussion).
- **Finite-sample convexity of the aggregation QP.** The population variance objective in (22) is convex because it is a variance of a linear combination of influence functions. In finite samples, separately estimated variance/covariance components can yield an approximately-convex (or slightly indefinite) quadratic form. Our implementation symmetrizes covariance inputs and adds a minimal ridge term when needed to ensure a well-posed convex optimization problem; the ridge is data-adaptive and vanishes as $N_t \rightarrow \infty$ under the stated conditions.
- **Fixed K vs. growing K .** The current theory treats the number of sources K as fixed. Allowing K to grow requires additional conditions to control accumulation of estimation error and selection (typically involving $\log K$ rates).

5 Federated Learning Algorithms

5.1 Two-Round Communication for Calibrated Estimation

This is the robust, standard procedure implementing the two-step estimation strategy.

1: Round 1: Initialization

1. **Source sites s_j compute and send:** An initial OR model parameter $\hat{\alpha}_{t,s_j}^{1,(0)}$ by minimizing Eq. (11).
2. **Target site t receives all $\hat{\alpha}_{t,s_j}^{1,(0)}$.** It also computes and sends back to each source site the summary statistic $\tilde{E}_t(\nabla_{\alpha} \psi(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^{1,(0)}))$, which is needed for the calibrated loss in Eq. (50).

2: Round 2: Calibrated Estimation

1. **Source sites s_j receive target info and compute:**
 - Calibrated parameters $\hat{\gamma}_{s_j,1}^{(1)}$ by minimizing the loss function from Eq. (50), using the received info and $\hat{\alpha}_{t,s_j}^{1,(0)}$.
 - Calibrated OR parameters $\hat{\alpha}_{t,s_j}^{1,(1)}$ by minimizing the loss function from Eq. (51) using the initial site/treatment model $\hat{\gamma}_{s_j,1}^{(0)}$ obtained from Eq. (12); this is the intended plug-in order in our two-level cross-fitting implementation.
 - The final correction term $\hat{\delta}_{t,s_j}^1$ and source variance $\hat{V}_{s_j}$ using the final parameters.
2. **Source sites s_j send back:** $\hat{\delta}_{t,s_j}^1$, N_{s_j} , $\hat{V}_{s_j}$, and the final $\hat{\alpha}_{t,s_j}^{1,(1)}$.
3. **Target site t computes final estimates:** For each s_j , it computes $\hat{\mu}_{t,s_j}^1$, $\hat{V}_{t,s_j}$, $\hat{C}_{ot,s_j}$, and all cross-site covariances $\hat{C}_{t,s_j,s_k}$. It also computes its target-only estimates $\hat{\mu}_{ot}^1$ and $\hat{V}_{ot}$.
4. **Target site t performs aggregation:** Solves the final optimization problem (22) for weights η .

6 Cross-Validation and Tuning Parameters

The procedure involves two distinct classes of regularization parameters. Those appearing in the nuisance models— λ_α and λ_γ —govern the sparsity of the outcome regression and calibration-weight coefficients, respectively, and their tuning controls the nuisance estimation rate. The aggregation penalty λ appearing in (22) plays a conceptually different role: it mediates the bias–variance trade-off between target-only estimation and source-assisted estimation, and its tuning determines which source sites contribute to the final estimator.

6.1 Cross-Validation for Nuisance Models

For each nuisance model (parameters γ and α), the regularization parameters $(\lambda_\gamma, \lambda_\alpha)$ are selected via K_f -fold cross-validation, where K_f is a fixed constant. The criterion for selection is the minimization of the corresponding calibrated loss function ((50) or (51)) on the validation folds. This is performed locally at each site during the federated step where the model is estimated.

6.2 Tuning the Aggregation Penalty

The regularization parameter λ in the aggregation objective (22) controls the bias–variance trade-off of the final estimator: small λ yields an unconstrained weighted combination (low bias, potentially high variance due to noisy weights), while large λ shrinks the weights towards zero, recovering the target-only estimator (higher bias, lower variance).

6.2.1 Data-Adaptive Lambda Grid

A fixed grid Λ (e.g. $[10^{-3}, 1]$) is not robust across problem scales because the “interesting” range of λ depends on the magnitudes of the variance components and the site discrepancies. To construct a data-adaptive grid, we identify the *natural scale* λ_{nat} at which the penalty term and the variance contribution are of comparable magnitude.

For source site s_j , the unpenalized variance contribution is

$$\sigma_j^2 := \frac{\widehat{V}_{t,s_j}}{N_t} + \frac{\widehat{V}_{s_j}}{N_{s_j}},$$

and the penalty scale is $d_j^2 := (\widehat{\mu}_{ot}^1 - \widehat{\mu}_{t,s_j}^1)^2$. Setting $\lambda d_j^2 = \sigma_j^2$ gives the transition point $\lambda_j = \sigma_j^2 / d_j^2$. When d_j^2 is very small (i.e. the source and target estimates nearly agree, suggesting low bias), we floor d_j^2 at σ_j^2 to avoid $\lambda_j \rightarrow \infty$:

$$\lambda_j = \frac{\sigma_j^2}{\max(d_j^2, \sigma_j^2)}.$$

The natural scale is defined as

$$\lambda_{\text{nat}} = \text{median}_{j=1,\dots,K} \lambda_j, \quad (23)$$

and the search grid spans two orders of magnitude on each side:

$$\Lambda = \{\lambda : \lambda \in [10^{-2} \lambda_{\text{nat}}, 10^2 \lambda_{\text{nat}}]\},$$

with $|\Lambda|$ evenly spaced on the log scale (e.g. 100 points). This ensures coverage of the full transition from under-regularised (η_j unconstrained) to over-regularised ($\eta_j \approx 0$), regardless of the absolute scale of the variance components and discrepancies.

6.2.2 Stabilised Selection via the One-Standard-Error Rule

Procedure. For each fold k_1 in the two-layer cross-fitting loop, the variance components $\widehat{V}_{ot}$, $\widehat{V}_{t,s_j}$, $\widehat{V}_{s_j}$, $\widehat{C}_{ot,s_j}$, and $\widehat{C}_{t,s_j,s_k}$ are already computed from disjoint evaluation data. For each candidate $\lambda \in \Lambda$:

1. Solve (22) for the optimal weights $\widehat{\eta}^{(\lambda)}$.
2. Evaluate the resulting aggregated variance $\widehat{\text{Var}}^{(\lambda)} = \widehat{\text{Var}}(\widehat{\mu}_{\text{agg}}^1; \widehat{\eta}^{(\lambda)})$ using the plug-in formula from (22).

Since the variance components are fixed for a given fold, no re-fitting of nuisance models is required; only the weight optimisation (a low-dimensional quadratic program over K variables) is repeated.

One-SE rule. Rather than selecting the minimiser $\lambda^* = \arg \min_{\lambda} \widehat{\text{Var}}^{(\lambda)}$ directly—which can overfit noisy variance component estimates—we adopt a conservative selection rule inspired by the one-standard-error (1-SE) rule of Hastie et al. [2009]:

$$\lambda_{1\text{SE}}^* = \max \left\{ \lambda \in \Lambda : \widehat{\text{Var}}^{(\lambda)} \leq (1 + \epsilon) \min_{\lambda' \in \Lambda} \widehat{\text{Var}}^{(\lambda')} \right\}, \quad (24)$$

where $\epsilon > 0$ is a relative tolerance (we use $\epsilon = 0.10$). This selects the *most regularised* λ whose aggregated variance is within ϵ of the minimum—i.e. the largest λ in the “flat” region of the variance curve.

Rationale. The classical 1-SE rule uses the standard error of cross-validated loss estimates across folds. Here, because λ only affects the weight optimisation (and not nuisance estimation), there are no replicate loss values from which to compute a standard error.¹ Instead, the relative tolerance ϵ serves as a pragmatic substitute: accepting a $\leq 10\%$ increase in estimated variance (equivalently, $\leq 4.9\%$ increase in SE) in exchange for more stable, regularised weights. When the variance curve is flat (common when site estimates are similar), this avoids selecting an arbitrarily small λ that produces extreme weights driven by estimation noise.

7 Simulation Study Design

7.1 Data Generation Framework

We conduct simulation studies to evaluate the performance of our proposed doubly robust estimators across various model specifications and sample sizes. Our simulation framework generates synthetic datasets with one target site and multiple source sites, allowing us to control the degree of heterogeneity between sites and assess robustness under model misspecification.

7.2 Covariate Generation and Transformation

7.2.1 Base Covariates

We generate n observations with p baseline covariates from a multivariate normal distribution:

$$\mathbf{X} = (X_1, \dots, X_p) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p) \quad (25)$$

where each component is truncated to the interval $[-2.5, 2.5]$ and subsequently standardized. In our simulations we study $p \in \{10, 50, 100, 200\}$ to assess performance across a range of covariate dimensions; the outcome and site models are sparse (at most 3 active covariates), so the remaining $p - 3$ coefficients are zero.

7.2.2 Nonlinear Covariate Transformations

To introduce model complexity and test robustness against misspecification, we create a transformed covariate matrix $\mathbf{X}^\dagger$. The first four covariates undergo nonlinear transformations:

$$X_1^\dagger = \text{scale}(\exp(0.5 \cdot X_1)) \quad (26)$$

$$X_2^\dagger = \text{scale} \left(10 + \frac{X_2}{1 + \exp(X_1)} \right) \quad (27)$$

$$X_3^\dagger = \text{scale}((0.04 \cdot X_1 \cdot X_3 + 0.5)^3) \quad (28)$$

$$X_4^\dagger = \text{scale}((X_2 + X_4 + 20)^2) \quad (29)$$

where $\text{scale}(\cdot)$ denotes standardization to zero mean and unit variance; the remaining covariates $X_j^\dagger = X_j$ for $j > 4$.

Our software implementation additionally supports milder transformations and the identity transformation (i.e., $\mathbf{X}^\dagger = \mathbf{X}$) as optional simulation settings. In the current simulation scripts, the default is the *mild* transformation; the stronger transformation above is used as a stress-test setting when specified.

¹In the `glmnet` setting, each fold provides an independent loss estimate, so the standard error of these K_f values quantifies selection uncertainty. In our setting, the variance components are computed once per fold and then shared across all λ evaluations, so fold-specific “loss” values are deterministically linked.

7.3 Site Membership and Treatment Assignment Model

7.3.1 Multinomial Logistic Site Assignment

We consider K source sites in addition to one target site. The joint probability of site membership and treatment assignment follows our proposed multinomial logistic model. For each source site s_j ($j = 1, 2, \dots, K$), we define site-specific parameters:

$$\gamma_{s_j,0} = (c_0, \epsilon_{j,0,1}, \epsilon_{j,0,2}, \mathbf{0}_{p-2}) \quad (30)$$

$$\gamma_{s_j,1} = (c_0, \epsilon_{j,1,1}, \epsilon_{j,1,2}, \mathbf{0}_{p-2}) \quad (31)$$

where $\epsilon_{j,a,k} \sim \mathcal{N}(0, \sigma_\epsilon^2)$ with $\sigma_\epsilon = 0.1$ represents small site-specific variations, and $c_0 = -0.15$ is a balancing intercept ensuring $P(R = t) \approx 1/(K + 1)$. Since the target site is the baseline category (all-zero γ), a negative intercept on source categories reduces their relative probability. The value is derived from $\mathbb{E}[\exp(\gamma^T \mathbf{Z})] = 1/2$ under unit-norm normalization with Gaussian covariates, yielding $P(R = t) = 1/(1 + 2K \cdot 1/2) = 1/(K + 1)$ regardless of K .

Implementation detail (normalization and intercept): In the simulation code, we first generate an *unnormalized* parameter vector

$$\tilde{\gamma}_{s_j,a} = (c_0, \epsilon_{j,a,1}, \epsilon_{j,a,2}, \mathbf{0}_{p-2}) \quad (32)$$

with the fixed intercept seed $c_0 = -0.15$, and then normalize (and optionally rescale) it as

$$\gamma_{s_j,a} = \kappa \cdot \tilde{\gamma}_{s_j,a} / \|\tilde{\gamma}_{s_j,a}\|_2, \quad (33)$$

where κ controls the shift strength (with $\kappa = 1$ matching the unit-norm constraint below). Consequently, the *normalized* intercept component of $\gamma_{s_j,a}$ is random and typically much more negative than c_0 ; the statement $c_0 = -0.15$ should be interpreted as a pre-normalization seed rather than the post-normalization intercept.

To see why a negative normalized intercept yields approximately balanced site sizes, note that when $\phi(\mathbf{X}) = (1, \mathbf{Z}^T)^T$ with $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\gamma = (\gamma_0, \gamma_{-0}^T)^T$,

$$\mathbb{E}[\exp\{\phi(\mathbf{X})^T \gamma\}] = \exp(\gamma_0 + \frac{1}{2} \|\gamma_{-0}\|_2^2). \quad (34)$$

Under the unit-norm constraint $\|\gamma\|_2 = 1$, we have $\|\gamma_{-0}\|_2^2 = 1 - \gamma_0^2$, and choosing γ_0 so that this expectation is near $1/2$ leads to a negative intercept (approximately $\gamma_0 \approx -0.84$ in the ideal Gaussian calculation). In finite samples with truncated/standardized covariates and random $\epsilon_{j,a,k}$, the balancing is approximate rather than exact.

Unit Norm Constraint: The parameter vectors are normalized such that:

$$\|\gamma_{s_j,0}\|_2 = 1 \quad \forall j = 1, 2, \dots, K \quad (35)$$

$$\|\gamma_{s_j,1}\|_2 = 1 \quad \forall j = 1, 2, \dots, K \quad (36)$$

7.3.2 Site Membership Probabilities

The probability of assignment to the target site is:

$$P(R = t | \phi(\mathbf{X})) = \frac{1}{1 + \sum_{j=1}^K (e^{\phi(\mathbf{X})^T \gamma_{s_j,0}} + e^{\phi(\mathbf{X})^T \gamma_{s_j,1}})} \quad (37)$$

where $\phi(\mathbf{X})$ represents either $\mathbf{X}$ or $\mathbf{X}^\dagger$ depending on the configuration.

The joint probabilities for source sites are:

$$P(R = s_j, A = a | \phi(\mathbf{X})) = \frac{e^{\phi(\mathbf{X})^T \gamma_{s_j,a}}}{1 + \sum_{k=1}^K (e^{\phi(\mathbf{X})^T \gamma_{s_k,0}} + e^{\phi(\mathbf{X})^T \gamma_{s_k,1}})} \quad (38)$$

7.3.3 Target Site Treatment Assignment

For subjects assigned to the target site, the treatment probability is derived from the aggregate source site behavior:

$$P(A = 1 | R = t, \phi(\mathbf{X})) = \frac{\sum_{j=1}^K P(R = s_j, A = 1 | \phi(\mathbf{X}))}{\sum_{j=1}^K [P(R = s_j, A = 0 | \phi(\mathbf{X})) + P(R = s_j, A = 1 | \phi(\mathbf{X}))]} \quad (39)$$

7.4 Outcome Generation Model

7.4.1 Generalized Outcome Model Structure

We specify generalized linear outcome models using the GLM framework. The conditional expectation of potential outcomes follows:

$$\mathbb{E}[Y(a) \mid \phi(\mathbf{X}), R = r] = \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}_r^a) \quad (40)$$

where $\psi(\cdot; \cdot)$ is the response function and $\boldsymbol{\alpha}_r^a$ are site-specific and treatment-specific parameters.

For the propensity score model, we similarly use the GLM framework where the linear predictor is:

$$\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j, a} \quad (41)$$

representing the linear predictor in the exponential family.

7.4.2 True Outcome Parameters

For binary outcomes, we use the logistic link function for the following two data-generating scenarios:

1. **Scenario 1: Homogeneous Outcome Models** In this scenario, target and source sites share the same outcome model parameters:

$$\boldsymbol{\alpha}_1 = (0.5, 0.5, 0.75, \mathbf{0}_{p-2})^T \quad (\text{treatment effect}) \quad (42)$$

$$\boldsymbol{\alpha}_0 = (-0.5, 0.75, 0, \mathbf{0}_{p-2})^T \quad (\text{control outcome}) \quad (43)$$

2. **Scenario 2: Heterogeneous Outcome Models** In this scenario, source sites have slightly different outcome model parameters than the target site:

- For target site:

$$\boldsymbol{\alpha}_1 = (0.5, 0.5, 0.75, \mathbf{0}_{p-2})^T \quad (\text{treatment effect}) \quad (44)$$

$$\boldsymbol{\alpha}_0 = (-0.5, 0.75, 0, \mathbf{0}_{p-2})^T \quad (\text{control outcome}) \quad (45)$$

- For source sites (mild heterogeneity):

$$\boldsymbol{\alpha}_1 = (0.5, 0.3, 0.45, \mathbf{0}_{p-2})^T \quad (\text{treatment effect}) \quad (46)$$

$$\boldsymbol{\alpha}_0 = (-0.5, 0.75, 0, \mathbf{0}_{p-2})^T \quad (\text{control outcome}) \quad (47)$$

7.4.3 Potential Outcomes and Observed Outcomes

Let $\phi(\mathbf{X})_i$ represent either $\mathbf{X}_i$ or $\mathbf{X}_i^\dagger$ depending on the configuration.

Binary outcome setting (default). We define

$$\mu_{i,a} := \psi(\phi(\mathbf{X})_i; \boldsymbol{\alpha}_a), \quad a \in \{0, 1\},$$

with logistic link $\psi(u) = \{1 + \exp(-u)\}^{-1}$, and generate the observed outcome as

$$Y_i \mid (A_i = a, \phi(\mathbf{X})_i) \sim \text{Bernoulli}(\mu_{i,a}).$$

Equivalently, under consistency, $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0)$ with $Y_i(a) \mid \phi(\mathbf{X})_i \sim \text{Bernoulli}(\mu_{i,a})$.

Continuous outcome setting (optional). For continuous outcomes, we use the identity link and additive Gaussian noise:

$$Y_i(1) = \phi(\mathbf{X})_i^T \boldsymbol{\alpha}_1 + \varepsilon_{i,1}, \quad (48)$$

$$Y_i(0) = \phi(\mathbf{X})_i^T \boldsymbol{\alpha}_0 + \varepsilon_{i,0}, \quad (49)$$

where $\varepsilon_{i,a} \sim \mathcal{N}(0, 0.01)$, and $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0)$.

7.5 Simulation Configurations

We consider three main configurations to assess model robustness:

Configuration C(1): *Correct Specification*

- Site assignment model uses $\mathbf{X}^\dagger$
- Outcome model uses $\mathbf{X}^\dagger$
- Both propensity score and outcome models are correctly specified

Configuration C(2): *Outcome Model Misspecification*

- Site assignment model uses $\mathbf{X}^\dagger$
- Outcome model uses $\mathbf{X}$ (misspecified)

Configuration C(3): *Propensity Score Model Misspecification*

- Site assignment model uses $\mathbf{X}$ (misspecified)
- Outcome model uses $\mathbf{X}^\dagger$

Configuration C(4): *Both Models Misspecified*

- Site assignment model uses $\mathbf{X}$ (misspecified)
- Outcome model uses $\mathbf{X}$ (misspecified)
- Tests the boundary of double robustness when neither model is correctly specified

We study the finite sample performance of our proposed **FACE-HD estimator** against four competing methods: (i) an estimator that leverages target data only (target-only), (ii) a sample-size adjusted estimator (SS), (iii) an inverse-variance weighted estimator (IVW), and (iv) an exponentially-tilted augmented inverse probability weighted estimator (Tilted-AIPW) that multiplies calibration-weighted site-specific AIPW estimators and aggregates via SS. We examine the bias, root mean square error (RMSE), and coverage probability of the 95% CIs of these estimators across 500 simulations.

8 Discussion

We have developed FACE-HD, a federated adaptive causal estimation framework designed for high-dimensional multi-site observational data. The framework combines an exponential-tilting calibration working model for the site/treatment-to-target density ratio, a two-step penalized calibrated estimation procedure embedded in two-level cross-fitting, and a penalized quadratic aggregation objective whose adaptive selection behavior prevents negative transfer. Under standard sparsity and regularity conditions, the resulting estimator of the target-site average treatment effect is $\sqrt{N_{\text{all}}}$ -consistent and asymptotically normal, and remains consistent whenever either the outcome model or the calibration weight model is correctly specified. A two-round federated protocol, together with a pragmatic single-round variant, preserves patient-level privacy by exchanging only summary statistics across sites.

Several aspects of the framework deserve further investigation. The present theory treats the number of source sites K as fixed; permitting K to grow with the total sample size requires uniform control of the nuisance estimation errors across sources and the variance/covariance components of the aggregated estimator, typically at $\log K$ rates. The GLM outcome and exponential-tilting calibration models can in principle be replaced with more flexible machine-learning nuisances within the same cross-fitted orthogonality architecture, at the cost of slower nuisance rates and more delicate truncation conditions. The current development focuses on a binary treatment observed at a single time point; continuous or multi-valued treatments and time-varying exposures are natural targets, although the calibration weight construction and the scale of the aggregation penalty would each require corresponding modification. Semi-supervised settings in which outcome labels are scarce at some sites can be incorporated by augmenting the calibrated losses with unlabeled covariate information, following a construction analogous to that of Hou et al. [2025]. Finally, an end-to-end open-source software implementation is distributed alongside this manuscript to facilitate reproducible research and transparent adoption in applied studies.

A One-Round Communication Algorithms for GLM Outcome Models

In settings where communication cost is a binding constraint, the two-round protocol of Section 5.1 can be streamlined into a single round of communication. The key observation is that the role of the source-specific initial outcome model in the cross-calibrated loss (50) is to provide a consistent plug-in estimator of the target-population outcome mean; this role can be served just as well by the target-only outcome model $\hat{\alpha}_{ot}^1$, which is available to the target site without any prior communication from the sources.

The resulting procedure substitutes $\hat{\alpha}_{ot}^1$ for the source-specific initial estimator throughout, thereby eliminating the inbound round of communication. This substitution introduces a pragmatic trade-off: the initialization is no longer adapted to each source site, which may inflate the finite-sample bias of the calibrated nuisance estimators when the target-only and source-specific initializers differ substantially. Under the alignment condition stated in the appendix of the calibrated loss derivation, however, the first-order asymptotic properties of the resulting pairwise and aggregated estimators coincide with those of the two-round procedure.

1: Round 1: Single Communication Round

1. Target site t computes and sends:

- Target-only outcome model parameters $\hat{\alpha}_{ot}^1$ by minimizing a standard regularized GLM loss on $\mathcal{D}_t$.
- The summary statistic $\tilde{E}_t(\nabla_{\alpha}\psi(\phi(\mathbf{X}); \hat{\alpha}_{ot}^1))$.
- The summary statistic $\tilde{E}_t[\phi(\mathbf{X})]$ for the initial site/treatment model in Eq. (12).
- The target site sends all quantities to all source sites s_j .

2. Source sites s_j receive info, compute, and send back:

- Using the received $\tilde{E}_t(\nabla_{\alpha}\psi(\cdot))$, $\tilde{E}_t[\phi(\mathbf{X})]$, and $\hat{\alpha}_{ot}^1$, each source site first computes an initial site/treatment model $\hat{\gamma}_{\text{init},s_j,1}$ via Eq. (12), then computes the calibrated $\hat{\gamma}_{s_j,1}$ via Eq. (50).
- Then, it finds its final outcome model $\hat{\alpha}_{t,s_j}^1$ by minimizing the calibrated loss in Eq. (51) using the initial site/treatment model $\hat{\gamma}_{\text{init},s_j,1}$ as plug-in.
- It computes the correction term $\hat{\delta}_{t,s_j}^1$ and source variance $\hat{V}_{s_j}$.
- Each source site s_j sends back to the target: $\hat{\delta}_{t,s_j}^1$, N_{s_j} , $\hat{V}_{s_j}$, and the final outcome model parameters $\hat{\alpha}_{t,s_j}^1$.

3. Target site t receives info and performs final aggregation:

- The target computes its target-only estimates $\hat{\mu}_{ot}^1$ and $\hat{V}_{ot}$.
- For each source site s_j , using the received $\hat{\alpha}_{t,s_j}^1$ and $\hat{\delta}_{t,s_j}^1$, it assembles the final estimator $\hat{\mu}_{t,s_j}^1$.
- It then computes all necessary target-side variance components: $\hat{V}_{t,s_j}$, $\hat{C}_{ot,s_j}$, and the cross-site covariances $\hat{C}_{t,s_j,s_k}$ for all pairs $j \neq k$.
- Finally, it solves the optimization problem in Eq. (22) to find the optimal aggregation weights $\boldsymbol{\eta}^*$ and compute the final estimate $\hat{\mu}_{\text{agg}}^1$.

B Two-Level Cross-fitting for Enhanced Robustness

To enhance robustness against the overfitting bias that can arise from the sequential estimation of nuisance parameters, we adapt the two-level cross-fitting strategy from [Hou et al., 2025]. This approach employs a nested sample-splitting scheme to ensure that the data used to train initial nuisance models is entirely disjoint from the data used for the subsequent calibration step. This not only mitigates bias but also relaxes the assumptions required for the initial estimators, achieving a property known as Neyman orthogonality. The following sections detail how this method is integrated into our one-round and two-round federated algorithms.

B.1 Calibrated Loss Functions

The core of the two-level cross-fitting method is the use of specially designed "calibrated" loss functions for refining the nuisance models. These losses are constructed to make the final ATE estimator insensitive to small estimation errors in the nuisance parameters. Following the principles in [Smucler et al., 2019] and [Tan, 2020], we introduce a calibration step after obtaining robust initial estimates.

Let $\hat{\alpha}_{\text{init}}$ and $\hat{\gamma}_{\text{init}}$ be the initial estimates of the outcome and site/treatment model parameters, respectively. To ensure numerical stability and satisfy theoretical positivity assumptions, we define a truncation function $\mathcal{T}(x) = \text{sign}(x) \min\{|x|, M_\tau\}$, where M_τ may depend on sample size and is chosen sufficiently large so truncation is asymptotically inactive at the population limit. (We avoid the symbol $\tau(\cdot)$ here to prevent collision with the ATE τ .) The calibrated losses for refining the parameters are then defined as follows. For exponential-family GLMs with canonical link, we use the standard notation that the negative log-likelihood has the form $-Y\theta + \Psi(\theta)$, where $\theta = \phi(\mathbf{X})^T \alpha$ is the linear predictor and $\psi(\theta) = \Psi'(\theta)$ is the conditional mean; $\psi'(\theta)$ denotes the derivative with respect to θ .

Calibrated Loss for Site/Treatment Model $\gamma_{s_j,1}$. The loss for $\gamma_{s_j,1}$ is weighted by a function of the initial outcome model prediction. For a GLM outcome model, this is:

$$\begin{aligned} \ell_{\gamma, \text{cal}}(\gamma_{s_j,1}; \hat{\alpha}_{\text{init}}) &= \tilde{E}_t [\nabla \alpha \psi(\phi(\mathbf{X}); \hat{\alpha}_{\text{init}})^T \gamma_{s_j,1}] \\ &\quad + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma_{s_j,1}} \psi'(\mathcal{T}(\phi(\mathbf{X})^T \hat{\alpha}_{\text{init}})) \right] \\ &\quad + \lambda_\gamma \|\gamma_{s_j,1}\|_1 \end{aligned} \quad (50)$$

Calibrated Loss for Outcome Model α_{t,s_j}^1 . Symmetrically, the loss for α_{t,s_j}^1 is a weighted GLM loss, where the weights depend on the initial site/treatment model prediction.

$$\begin{aligned} \ell_{\alpha, \text{cal}}(\alpha_{t,s_j}^1; \hat{\gamma}_{\text{init}}) &= \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \hat{\gamma}_{\text{init}})}} \left(-Y \cdot (\phi(\mathbf{X})^T \alpha_{t,s_j}^1) + \Psi(\phi(\mathbf{X})^T \alpha_{t,s_j}^1) \right) \right] \\ &\quad + \lambda_\alpha \|\alpha_{t,s_j}^1\|_1 \end{aligned} \quad (51)$$

Asymptotic Alignment Remark. In the two-level algorithm, the calibrated γ -loss uses $\hat{\alpha}_{\text{init}}$ while the calibrated α -loss uses $\hat{\gamma}_{\text{init}}$. In the appendix proofs, we assume these initial estimators converge to the same population nuisance targets as the final calibrated estimators, so this plug-in order does not change the first-order asymptotic results.

B.2 Two-Round Algorithm with Two-Level Cross-fitting

This algorithm embeds the two-level cross-fitting logic within the two-round communication protocol. While it increases the computational load, it provides strong theoretical guarantees against overfitting bias.

1. **Initialization:** At each site $R \in \{t, s_1, \dots, s_K\}$, randomly partition the local data $\mathcal{D}_R$ into K_f disjoint, equally sized folds: $\{\mathcal{D}_{R,k}\}_{k=1}^{K_f}$. The following steps describe the computation for a single main estimation fold $k_1 \in \{1, \dots, K_f\}$.
2. **Round 1: Initial Nuisance Models**

1. **Source sites s_j compute and send:** For each secondary fold $k_2 \in \{1, \dots, K_f\} \setminus \{k_1\}$, source site s_j computes:

- An initial outcome model parameter $\hat{\alpha}_{\text{init},(k_1,k_2)}^1$ by minimizing the standard GLM loss from Equation 11 using all its local data **except** folds k_1 and k_2 , i.e., on data $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$.

Note: The initial site/treatment model $\hat{\gamma}_{\text{init}}$ is **not** computed here as it requires target site information.

It sends the set of $K_f - 1$ initial outcome models $\{\hat{\alpha}_{\text{init},(k_1,k_2)}^1\}_{k_2 \neq k_1}$ to the target site.

2. **Target site t receives and sends back:** For each source s_j and each secondary fold $k_2 \neq k_1$, the target computes two summary statistics using only data from its local fold $\mathcal{D}_{t,k_2}$:

- For the *calibrated* site/treatment model: $\tilde{E}_{t,k_2}[\nabla_{\alpha}\psi(\phi(\mathbf{X}); \hat{\alpha}_{\text{init},(k_1,k_2)}^1)]$.
- For the *initial* site/treatment model: $\tilde{E}_{t,k_2}[\phi(\mathbf{X})]$.

It sends the corresponding set of $K_f - 1$ pairs of summaries back to source site s_j .

3: Round 2: Calibrated Nuisance Models and Estimation

1. Source sites s_j receive target info and compute:

- For each secondary fold $k_2 \neq k_1$, the source computes initial and then calibrated parameters:
 - **First**, compute the initial site/treatment model $\hat{\gamma}_{\text{init},(k_1,k_2)}$ by minimizing its standard loss (Eq. 12) using the received summary $\tilde{E}_{t,k_2}[\phi(\mathbf{X})]$ and its own data from $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$.
 - **Then**, compute the calibrated parameters on its local data from fold $\mathcal{D}_{s_j,k_2}$:
 - * $\hat{\gamma}_{\text{cal},(k_1,k_2)}$ by minimizing the calibrated loss (50) using its initial outcome model $\hat{\alpha}_{\text{init},(k_1,k_2)}^1$ and the summary $\tilde{E}_{t,k_2}[\nabla_{\alpha}\psi(\cdot)]$.
 - * $\hat{\alpha}_{\text{cal},(k_1,k_2)}^1$ by minimizing the calibrated loss (51) using the initial site/treatment model $\hat{\gamma}_{\text{init},(k_1,k_2)}$ computed just above.
- It computes the final nuisance models for the main fold k_1 by averaging the calibrated parameters from the $K_f - 1$ secondary folds:

$$\hat{\alpha}_{t,s_j,(k_1)}^1 := \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \hat{\alpha}_{\text{cal},(k_1,k_2)}^1, \quad \hat{\gamma}_{s_j,(k_1)} := \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \hat{\gamma}_{\text{cal},(k_1,k_2)}$$

- Using these final averaged parameters, it computes the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$ and source variance $\hat{V}_{s_j,(k_1)}$ on its own main estimation data from fold $\mathcal{D}_{s_j,k_1}$.
2. **Source sites s_j send back:** The final averaged parameters $\hat{\alpha}_{t,s_j,(k_1)}^1$, the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$, and variance component $\hat{V}_{s_j,(k_1)}$.
 3. **Target site t computes fold-specific estimate:** Using the received information and its own data, the target computes:

- *Inner-fold variance components:* For each secondary fold $k_2 \neq k_1$, using the per- (k_1, k_2) calibrated parameters (available from Round 2) and inner-fold target-only models (trained on target data from folds $\setminus \{k_1, k_2\}$), compute all variance-covariance components on data from fold $\mathcal{D}_{k_2}$. Average these across the $K_f - 1$ inner folds to obtain $\hat{V}^{(-k_1)}$, $\hat{C}_{\text{ot}}^{(-k_1)}$, and $\hat{C}_{\text{cross}}^{(-k_1)}$.
- *Weight optimization:* Solve (22) using these inner-fold variance components to get $\hat{\eta}^{(-k_1)}$.

- *Estimation:* Using the averaged nuisances $\hat{\vartheta}^{(-k_1)}$ and the weights $\hat{\eta}^{(-k_1)}$, compute the fold-specific aggregated estimate $\hat{\mu}_{\text{agg},k_1}^1$ and per-sample IF values on the evaluation fold $\mathcal{D}_{k_1}$.

4. **Final Aggregation:** The entire process is repeated for each main fold $k_1 = 1, \dots, K_f$. The final point estimate is the average: $\hat{\mu}_{\text{agg}}^1 = \frac{1}{K_f} \sum_{k_1=1}^{K_f} \hat{\mu}_{\text{agg},k_1}^1$.

Inner M -fold weight estimation. To ensure that the aggregation weights $\hat{\eta}^{(-k_1)}$ are independent of the evaluation data $\mathcal{D}_{k_1}$, we estimate the variance–covariance components for weight optimisation (Eq. (22)) via an inner cross-fitting loop within the training set $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$. Specifically, we set $M = K_f - 1$ inner folds that coincide with the secondary folds $\{k_2 : k_2 \neq k_1\}$. For each inner fold k_2 :

1. Use the per- (k_1, k_2) nuisance estimates $\hat{\vartheta}^{(-k_1, -k_2)}$ —trained on data excluding both $\mathcal{D}_{k_1}$ and $\mathcal{D}_{k_2}$ —and the corresponding inner-fold target-only estimates (trained on target data from the same reduced set) to compute IF components $(\hat{\varphi}_{\text{ot}}, \hat{\zeta}, \hat{\xi})$ on the inner evaluation fold $\mathcal{D}_{k_2}$.
2. Compute all variance–covariance components $(\hat{V}_{\text{ot}}, \hat{V}_t, \hat{V}_s, \hat{C}_{\text{ot}}, \hat{C}_{\text{cross}})$ from these IF components on $\mathcal{D}_{k_2}$.

The components are then averaged across the $K_f - 1$ inner folds:

$$\hat{V}_{\cdot}^{(-k_1)} := \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \hat{V}_{\cdot, (k_1, k_2)},$$

and similarly for $\hat{C}_{\text{ot}}^{(-k_1)}$ and $\hat{C}_{\text{cross}}^{(-k_1)}$. The weights $\hat{\eta}^{(-k_1)}$ are then obtained by solving (22) with these inner-fold variance components. Because the entire computation uses only data from $\mathcal{T}_{k_1}$ (with out-of-inner-fold nuisances), $\hat{\eta}^{(-k_1)}$ is independent of the evaluation fold $\mathcal{D}_{k_1}$.

Sample-level variance (unified estimating equation; multi-sample regime). The point estimate and variance are derived from the *same* set of per-observation uncentered influence functions. Under the standard multi-site sampling regime (independent samples per site with fixed sampling fractions), the variance of the overall mean depends on *within-site* variation only. For each observation i in evaluation fold $\mathcal{D}_{k(i)}$, define

$$\hat{\Phi}_i := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \hat{\eta}_j^{(-k(i))}) \tilde{Y}_{\text{ot},i} + \sum_j \hat{\eta}_j^{(-k(i))} \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) \right], & i \in \text{target}, \\ \frac{N_{\text{all}}}{N_{s_j}} \hat{\eta}_j^{(-k(i))} w(\mathbf{X}_i; \hat{\gamma}_{s_j,1}^{(-k(i))}) (Y_i - \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))})), & i \in \text{source } s_j, \end{cases}$$

Here $N_{\text{all}} = N_t + \sum_j N_{s_j}$, $\tilde{Y}_{\text{ot},i}$ is the *uncentered* AIPW pseudo-outcome for the target-only estimator,

$$\tilde{Y}_{\text{ot},i} := \frac{A_i Y_i}{\hat{\pi}_{\text{ot}}(\mathbf{X}_i)} - \left(\frac{A_i}{\hat{\pi}_{\text{ot}}(\mathbf{X}_i)} - 1 \right) \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{\text{ot}}^{1,(-k(i))}),$$

so that the centered IF defined in (13) satisfies $\hat{\varphi}_{\text{ot},i} = \tilde{Y}_{\text{ot},i} - \hat{\mu}_{\text{ot},k(i)}^1$ (the fold-specific target-only point estimate). Likewise, the centered target-site component (14) satisfies $\hat{\zeta}_{t,s_j,i} = \psi(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^{1,(-k(i))}) - \tilde{E}_{t,k(i)}[\psi(\cdot)]$. All nuisance parameters $\hat{\alpha}_{t,s_j}^{1,(-k)}$, $\hat{\gamma}_{s_j,1}^{(-k)}$, and $\hat{\pi}_{\text{ot}}^{(-k)}$ are trained on data excluding the evaluation fold $\mathcal{D}_k$ (with the inner M -fold calibration/averaging as described above). Note that the source contribution is written in uncentered form; subtracting the fold-specific correction term $\hat{\delta}_{t,s_j}^{(-k)} = \tilde{E}_{s_j,k}[w(Y - \psi)]$ yields the centered influence component $\hat{\xi}_{t,s_j,i}$ defined in (15).

The point estimate and variance are then derived from the *same* set $\{\hat{\Phi}_i\}_{i=1}^{N_{\text{all}}}$:

$$\hat{\mu}_{\text{agg}}^1 = \bar{\Phi} := \frac{1}{N_{\text{all}}} \sum_{i=1}^{N_{\text{all}}} \hat{\Phi}_i,$$

$$\widehat{\text{Var}}(\hat{\mu}_{\text{agg}}^1) = \frac{1}{N_{\text{all}}^2} \sum_{g \in \{t, s_1, \dots, s_K\}} \sum_{i \in \mathcal{D}_g} (\hat{\Phi}_i - \bar{\Phi}_g)^2, \quad \bar{\Phi}_g := \frac{1}{N_g} \sum_{i \in \mathcal{D}_g} \hat{\Phi}_i.$$

This within-site centered form is the natural plug-in estimator for the variance of a mean under independent multi-sample (site-stratified) sampling; global centering would add an ANOVA between-site term that is not sampling variability in this regime. With balanced folds, $\bar{\Phi}$ coincides with the fold-average $(1/K_f) \sum_k \hat{\mu}_{\text{agg},k}^1$.

B.3 One-Round Algorithm with Two-Level Cross-fitting

This algorithm adapts the two-level split to the more communication-efficient one-round protocol. The target site bootstraps the entire cross-fitting process by computing and sending all necessary initial models and summary statistics in a single round.

- 1: **Initialization:** All sites partition their local data into K_f folds: $\{\mathcal{D}_{R,k}\}_{k=1}^{K_f}$. The following describes the computation for a single main estimation fold k_1 .

- 2: **Round 1: Single Communication Round**

1. **Target site t computes and sends:**

- For each secondary fold k_2 where $k_2 \neq k_1$:
 - Computes a preliminary target-only outcome model $\hat{\alpha}_{\text{init},(k_1,k_2)}^1$ on its local data from all folds **except** k_1 and k_2 .
 - Computes the corresponding summary statistic for the *calibrated* model:

$$\tilde{E}_{t,k_2}[\nabla_{\alpha}\psi(\phi(\mathbf{X}); \hat{\alpha}_{\text{init},(k_1,k_2)}^1)]$$
 on its local fold $\mathcal{D}_{t,k_2}$.
 - Computes the summary statistic for the *initial* model: $\tilde{E}_{t,k_2}[\phi(\mathbf{X})]$ on its local fold $\mathcal{D}_{t,k_2}$.
- It sends the full set of $(K_f - 1)$ initial models and their corresponding $(K_f - 1)$ pairs of summary statistics (all associated with main fold k_1) to all source sites.

2. **Source sites s_j receive info, compute, and send back:**

- For each secondary fold $k_2 \neq k_1$, the source site computes its initial and calibrated parameters:
 - **First**, compute the initial site/treatment model $\hat{\gamma}_{\text{init},(k_1,k_2)}$ on its local data from $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j,k_1} \cup \mathcal{D}_{s_j,k_2})$ using the received summary $\tilde{E}_{t,k_2}[\phi(\mathbf{X})]$.
 - **Then**, using the received information from the target and its own initial models, it computes the calibrated parameters $\hat{\gamma}_{\text{cal},(k_1,k_2)}$ and $\hat{\alpha}_{\text{cal},(k_1,k_2)}^1$ on its local data from fold $\mathcal{D}_{s_j,k_2}$.
- It averages these to get the final nuisance models for the main fold k_1 : $\hat{\alpha}_{t,s_j,(k_1)}^1$ and $\hat{\gamma}_{s_j,(k_1)}$.
- It computes the correction term $\hat{\delta}_{t,s_j,(k_1)}^1$ and variance $\hat{V}_{s_j,(k_1)}$ on its main estimation data from fold $\mathcal{D}_{s_j,k_1}$.
- Each source sends back the final averaged parameters $\hat{\alpha}_{t,s_j,(k_1)}^1$ along with $\hat{\delta}_{t,s_j,(k_1)}^1$ and $\hat{V}_{s_j,(k_1)}$.

3. **Target site t receives info and computes fold-specific estimate:**

- Using the received per- (k_1, k_2) calibrated parameters and inner-fold target-only models, the target computes variance-covariance components on each inner fold $\mathcal{D}_{k_2}$ and averages them across $k_2 \neq k_1$ to obtain the inner-fold variance estimates.
- It solves the weight optimisation using these inner-fold variance components and computes the fold-specific aggregated estimate $\hat{\mu}_{\text{agg},k_1}^1$ and per-sample IF values on the evaluation fold $\mathcal{D}_{k_1}$.

- 3: **Final Aggregation:** As before, the process is repeated for each main fold $k_1 = 1, \dots, K_f$. The point estimate is $\hat{\mu}_{\text{agg}}^1 = \frac{1}{K_f} \sum_{k_1} \hat{\mu}_{\text{agg},k_1}^1$, and the variance is computed using the sample-level DML formula with inner M -fold weights, as described in the two-round algorithm above.

C One-Round Communication Algorithm for Linear Outcome Models

When the outcome models are linear, i.e., $\psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \boldsymbol{\alpha}^T \phi(\mathbf{X})$, the framework simplifies significantly. Specifically, the gradient of the outcome model, $\nabla_{\boldsymbol{\alpha}} \psi(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \phi(\mathbf{X})$, does not depend on the parameter $\boldsymbol{\alpha}$. This breaks the iterative dependency between the outcome and site/treatment models, allowing for a highly efficient one-round estimation at the source sites. The strategy is for the target site to send sufficient statistics that allow source sites to locally perform a sequential estimation.

C.1 Derivations for One-Round Linear Algorithm

C.1.1 Variance and Covariance

Here we first derive the key formulas that allow variance and covariance terms involving target site data to be computed efficiently using pre-calculated sufficient statistics in the linear model case. For notational simplicity, we write $\mathbf{X}$ in place of $\phi(\mathbf{X})$ throughout this section.

Derivation of Target-Side Variance $\hat{V}_{t,s_j}$. The target-side variance for the estimator from source s_j is the variance of its influence function component on the target data. By definition, and noting that the mean of the influence function component is zero:

$$\hat{V}_{t,s_j} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i})^2 \quad (52)$$

For a linear model, $\psi(\mathbf{X}_i; \boldsymbol{\alpha}) = \boldsymbol{\alpha}^T \mathbf{X}_i$. The influence function component is:

$$\hat{\zeta}_{t,s_j,i} = \psi(\mathbf{X}_i; \hat{\boldsymbol{\alpha}}_{t,s_j}^1) - \tilde{E}_t[\psi(\mathbf{X}; \hat{\boldsymbol{\alpha}}_{t,s_j}^1)] = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \mathbf{X}_i - \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \bar{\mathbf{X}}_t = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) \quad (53)$$

Squaring this term gives a scalar, which can be expressed as a quadratic form:

$$(\hat{\zeta}_{t,s_j,i})^2 = \left(\hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) \right) \left((\mathbf{X}_i - \bar{\mathbf{X}}_t)^T \hat{\boldsymbol{\alpha}}_{t,s_j}^1 \right) = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) (\mathbf{X}_i - \bar{\mathbf{X}}_t)^T \hat{\boldsymbol{\alpha}}_{t,s_j}^1 \quad (54)$$

Summing over all $i \in \mathcal{D}_t$ and dividing by N_t , the central term becomes the sample covariance matrix of the covariates at the target site, $\hat{\Sigma}_t$:

$$\hat{V}_{t,s_j} = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \left(\frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\mathbf{X}_i - \bar{\mathbf{X}}_t) (\mathbf{X}_i - \bar{\mathbf{X}}_t)^T \right) \hat{\boldsymbol{\alpha}}_{t,s_j}^1 = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \hat{\Sigma}_t \hat{\boldsymbol{\alpha}}_{t,s_j}^1 \quad (55)$$

Derivation of Covariance Term $\hat{C}_{ot,s_j}$. This term is the covariance between the target-only influence function and the target-side component from source s_j .

$$\hat{C}_{ot,s_j} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot}) (\hat{\zeta}_{t,s_j,i}) \quad (56)$$

$$= \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot}) \left(\hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) \right) \quad (57)$$

Since $\hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T}$ is constant with respect to the sum, we can factor it out:

$$\hat{C}_{ot,s_j} = \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \left(\frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\varphi}_{ot,i} - \bar{\varphi}_{ot}) (\mathbf{X}_i - \bar{\mathbf{X}}_t) \right) \quad (58)$$

$$= \hat{\boldsymbol{\alpha}}_{t,s_j}^{1,T} \hat{\mathbf{C}}_{ot,X} \quad (59)$$

where $\hat{\mathbf{C}}_{ot,X}$ is the pre-computed vector of covariances between the target-only IF and the covariates.

Derivation of Cross-Site Covariance Term $\hat{C}_{t,s_j,s_k}$. This term is needed by the target site for the final aggregation. It requires the parameter vectors from two different source sites, s_j and s_k .

$$\hat{C}_{t,s_j,s_k} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\hat{\zeta}_{t,s_j,i})(\hat{\zeta}_{t,s_k,i}) \quad (60)$$

$$= \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} \left(\hat{\alpha}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) \right) \left(\hat{\alpha}_{t,s_k}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) \right) \quad (61)$$

Rewriting the second scalar term as its transpose allows us to form the quadratic expression:

$$\hat{C}_{t,s_j,s_k} = \frac{1}{N_t} \sum_{i \in \mathcal{D}_t} \hat{\alpha}_{t,s_j}^{1,T} (\mathbf{X}_i - \bar{\mathbf{X}}_t) (\mathbf{X}_i - \bar{\mathbf{X}}_t)^T \hat{\alpha}_{t,s_k}^1 \quad (62)$$

$$= \hat{\alpha}_{t,s_j}^{1,T} \left(\frac{1}{N_t} \sum_{i \in \mathcal{D}_t} (\mathbf{X}_i - \bar{\mathbf{X}}_t) (\mathbf{X}_i - \bar{\mathbf{X}}_t)^T \right) \hat{\alpha}_{t,s_k}^1 = \hat{\alpha}_{t,s_j}^{1,T} \hat{\Sigma}_t \hat{\alpha}_{t,s_k}^1 \quad (63)$$

This shows that once the target has $\hat{\Sigma}_t$ and receives the parameter vectors $\hat{\alpha}_{t,s_j}^1$ and $\hat{\alpha}_{t,s_k}^1$, it can compute this final covariance term.

C.1.2 Loss Functions

For the linear case, the general loss functions from the main text specialize to the following forms. This allows for a sequential estimation procedure: first γ is estimated to balance the raw covariates $\phi(\mathbf{X})$, and then α is estimated using weights derived from γ .

Loss Function for Calibration (Tilting) Model $\gamma_{s_j,1}$. The loss for γ becomes a direct covariate balancing objective using an exponential tilting formulation. As it only depends on the features $\phi(\mathbf{X})$, it can be minimized first without knowledge of α .

$$\ell(\gamma_{s_j,1}) = \left(\tilde{E}_t[\phi(\mathbf{X})] \right)^T \gamma_{s_j,1} + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma_{s_j,1}} \right] + \lambda_\gamma \|\gamma_{s_j,1}\|_1 \quad (64)$$

The term $\tilde{E}_t(\phi(\mathbf{X}))$ is the mean covariate vector $\bar{\phi}_t$ sent from the target site.

Loss Function for Outcome Model α_{t,s_j}^1 . The outcome model parameters are estimated using weighted least squares (Lasso Regression), which is the specific form of the general GLM loss for linear models. The weights are derived from the calibration/tilting model estimated in the previous step.

$$\ell(\alpha_{t,s_j}^1) = \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j,1}}} (Y - \alpha_{t,s_j}^{1,T} \phi(\mathbf{X}))^2 \right] + \lambda_\alpha \|\alpha_{t,s_j}^1\|_1 \quad (65)$$

C.2 Algorithm Steps

1: Round 1: Single Communication Round

1. Target site t computes and sends:

- Target-only estimates: $\hat{\alpha}_{ot}^1$, $\hat{\mu}_{ot}^1$, and $\hat{V}_{ot}$.
- Sufficient statistics from $\mathcal{D}_t$: the covariate mean $\bar{\phi}_t$, the covariate covariance matrix $\hat{\Sigma}_t$, the sample size N_t , and the covariance vector $\hat{C}_{ot,X}$.
- The target site broadcasts this package of information to all source sites.

2. Source sites s_j receive info, compute, and send back:

- Each source site first computes its density ratio parameters $\hat{\gamma}_{s_j,1}$ by minimizing Eq. (64) using the received $\bar{\phi}_t$.
- Then, it computes its outcome model parameters $\hat{\alpha}_{t,s_j}^1$ by minimizing the weighted least squares loss in Eq. (65), using weights derived from the newly computed $\hat{\gamma}_{s_j,1}$.

- Using the received statistics, it computes the final estimator $\hat{\mu}_{t,s_j}^1$, the target-side variance $\hat{V}_{t,s_j} = \hat{\alpha}_{t,s_j}^{1,T} \hat{\Sigma}_t \hat{\alpha}_{t,s_j}^1$, and the covariance $\hat{C}_{ot,s_j} = \hat{C}_{ot,X}^T \hat{\alpha}_{t,s_j}^1$.
- Using its own data, it computes the source-side variance $\hat{V}_{s_j}$.
- Each source site s_j sends back to the target: $\hat{\mu}_{t,s_j}^1$, $\hat{V}_{t,s_j}$, $\hat{V}_{s_j}$, $\hat{C}_{ot,s_j}$, N_{s_j} , and the parameter vector $\hat{\alpha}_{t,s_j}^1$ (needed for cross-site covariances).

3. Target site t receives info and performs final aggregation:

- For each pair of source sites (s_j, s_k) , the target computes the cross-site covariance: $\hat{C}_{t,s_j,s_k} = \hat{\alpha}_{t,s_j}^{1,T} \hat{\Sigma}_t \hat{\alpha}_{t,s_k}^1$.
- With all estimators and variance-covariance components now available, the target solves the optimization problem in Eq. (22) to find the optimal aggregation weights $\boldsymbol{\eta}^*$ and compute the final estimate $\hat{\mu}_{\text{agg}}^1$.

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